



Three good links, one edge each

Second Edition, Short Run and Even Keel against each other and against today's link — the figures, then a recommendation

What all three share

Ten designs were built and checked. These three have no all-round better alternative in the pack; the one design that beats them on anything, Long Break, pays for it elsewhere and is on page 2. On everything below the three are the same, and on each line better than or level with today — with one exception, leave, which is under the costs on page 2.

✓ All **9 of 9** December staffing rules, and all 3 flexible ones (today 4 of 9, 0 flexible)

✓ **7 full weekends off in 26**, never more than 5 weeks apart (today 4 in 20, up to 7 apart)

✓ Every closing shift **30 to 45 minutes shorter**; no shift over 9h 00m (today 9h 10m)

✓ **No fatigue warning**, fixed rota or with a cover week placed badly (today 4, up to 5)

✓ The same **35-hour** week and **218.6 days** a year, under the ceiling of 219 (today 219.0)

✓ **4 single rest days** in the rotation, as today; rest days otherwise in pairs or longer (22 of 26 breaks, today 13 of 17)

Where they differ

The best of the three on each line is in bold green. Today is shown for scale, not ranked.

MEASURE	TODAY	SECOND EDITION	SHORT RUN	EVEN KEEL
Most days in a row, fixed rota	7	6	5	5
Most days in a row, cover week placed badly	9	6	6	6
Heaviest week	43h 50m	42h 00m	42h 30m	41h 30m
Lightest week	25h 10m	25h 30m	24h 30m	24h 30m
Shortest rest between shifts	12h 30m	14h 20m	14h 20m	14h 35m
Weeks on one turn, Monday to Friday	7 of 16	18 of 21	18 of 21	18 of 21
Weeks mixing earlies and lates	7	0	0	1
Week-to-week change of start time	4h 00m	1h 01m	1h 13m	1h 16m
Finishing at 23:00 or later, each a year	39	42	42	44
Shift times to learn that are new	0 of 18	6 of 15	6 of 15	7 of 15
Fit to the trains, weekday · Sat · Sun (0 = staff track the trains exactly)	52.0 · 65.9 · 71.7	29.1 · 12.7 · 29.0	29.1 · 12.7 · 29.0	29.4 · 14.5 · 29.0

Each one's edge

Second Edition

SE-26-F1 · dea6417f

The steadiest. One shift time Monday to Friday in 18 of 21 weeks and the gentlest week-to-week change of the three, 1h 01m. Its cost is the one everybody else has shed: 6 days in a row on the fixed rota.

Short Run

SR-26-F1 · 618348d6

The shortest stretches. Second Edition's rota, same times and same weeks on one turn, with no run over 5 days on the fixed rota. The price is one week at 42h 30m (Second Edition 42h 00m) and a slightly rougher step.

Even Keel

EK-26-H1 · 8e9a1bcf

The lightest weeks and the longest rests. A different duty table: no week over 41h 30m, shortest rest 14h 35m, no run over 5. It pays with 44 late finishes a year (the others 42), 7 new times and 1 mixed week.

How they were made, and how they were checked

Second Edition came from a search over the rest-day layout first and the shift times second, under every rule a finished rota must meet. Short Run is the same duty table on a layout an exact solver proved to exist (no run over five days with still only four single rest days — the annealing search had missed it in sixteen attempts). Even Keel is an outside reviewer's reordering of Second Gear, verified here by the app's own checks before it shipped. Every figure here, on the sheets and in the presentations is computed from the rota by the same code, and four independent audits recomputed them from the grids and found no wrong number; the wording around them has been corrected where those audits faulted it.

Against today, and a recommendation

Where the three agree they share a column — which on most of what a colleague feels, they do

Against today's link

MEASURE	TODAY	SECOND EDITION	SHORT RUN	EVEN KEEL
Full weekends off, in the rotation	4 in 20		7 in 26	
Full weekends off, about a year	10		14	
Longest wait between full weekends	7 weeks		5 weeks	
Contracted days at work a year	219.0		218.6	
Saturdays a year, each	26		28	
Sunday overtime duties a year, each	21		20	
Starting at 06:20, each a year	62		60	
Average shift, Monday to Saturday	8h 14m		8h 16m	
Longest shift	9h 10m		9h 00m	
14 days' leave buys, at best	30 days off		28 days off	
14 days' leave buys, at worst	19 days off		20 days off	
Leave for four full weeks off	14 days		15 days	

Per-year figures are the rotation's count scaled to 52 weeks; days a year is Monday to Saturday, a cover week counting four. Sundays stay overtime.

Why these three and not the other seven

Ten designs are in the pack. Six of the others are beaten line for line by one of these three — the same table with more mixed weeks, a rougher step, a heavier week or a fifth single rest day — so presenting them would add a choice without adding an option. The seventh, Long Break, alone brings four weeks off back to 14 days' leave, for a fifth single rest day, two fewer one-turn weeks and a 1h 44m step; it stays in the pack as the leave-first option.

The honest costs, shared by all three

- **Late finishes.** 42 a year each at 23:00 or later (today 39), 44 on Even Keel. It is the December staffing: three to the close every day where today's weekdays have two, so no link meeting the rules can have fewer than 42.
- **Leave.** The best 14-day stretch is 28 days off, not today's 30; four full weeks off takes 15 days' leave, not 14. The worst stretch is a day better. Say this early: it is the one line where every proposal is worse than today on something people plan a year around.
- **Saturdays.** About 2 more a year each (28, today 26) — the price of 7 full weekends spaced 5 apart instead of 4 spaced up to 7.
- **New times.** 6 of 15 shift times are new on Second Edition and Short Run (9 are worked today), 7 of 15 on Even Keel. Four of the new ones are today's lates starting later — two of them the closers — so shorter.

Recommendation

Short Run. On the fixed rota — the worst case with a cover week placed badly is six days for all three — it is Second Edition with the last six-day stretch taken out — 6 days in a row on Second Edition's fixed rota, never more than 5 on Short Run's — and what it gives up is a half hour on one week (42h 30m against 42h 00m, both under today's 43h 50m) and 0h 12m on the week-to-week step. A sixth day in a row is something a colleague feels and remembers; thirty minutes once in 26 weeks is not. That is the right direction for a staff proposal.

But the choice is closer than the table above makes it look. Weight every line and the three finish within about a point of a hundred, and which is first turns on whether the run is measured on the fixed rota or on a cover week placed badly. So choose by the edge that matters here:

- **Choose Short Run** if days in a row is what colleagues raise first.
- **Choose Even Keel** if the heaviest week and the shortest rest matter more than two late finishes a year and a seventh new time; those two late finishes buy a fourth person to the Saturday close, which a manager may count as a gain. It is the only one on a different duty table, so it is a genuine alternative rather than a variant.
- **Choose Second Edition** if steady shift times week to week carry the day: the gentlest week-to-week change of the three, at the price of the six-day stretch. It is Short Run's fallback, already on the shelf.

The caveat all three carry: the run and fatigue figures are for the duties the rota fixes. A cover week's four duties are placed later by the roster clerk, and placed badly they can make six days in a row on any of them — the sheets print both figures.